

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

BIOMOLECULES

LECTURE-3

ZOOLOGY

DAVOOD SIR (CEO of NG)

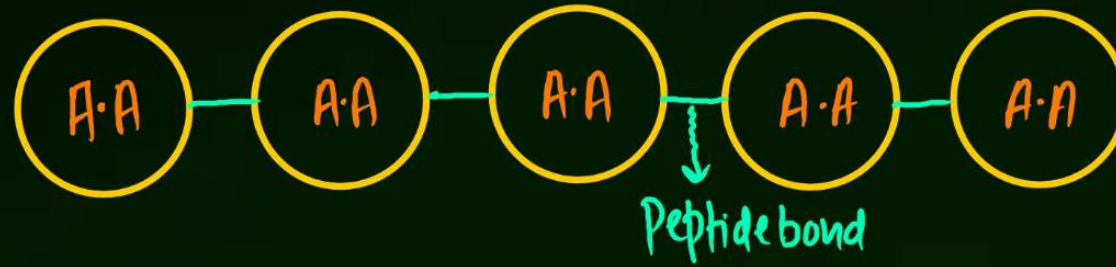


Physics Wallah

Topics To Be Covered

- 1 **Proteins and Levels of Proteins**
- 2 **Previous Year NEET MCQ's**
- 3

Proteins



Definition

→ Protein is a heteropolymer of 20-different amino acids. At least a protein should contain 50 different amino acids linked by peptide bond.

- Protein can never be a homopolymer [same type of amino acids].

- Most abundant protein
 - Animals → Collagen protein
 - Whole Biosphere → RuBisCO protein.

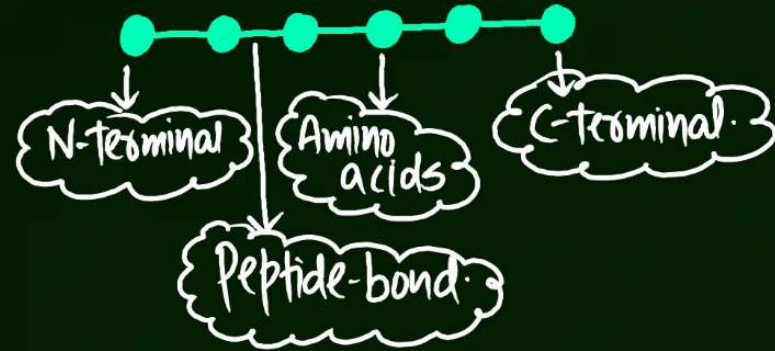
Plants.
↓
Ribulose Bisphosphate Carboxylase Oxygenase.

Structure of Proteins Levels of proteins



Structures of Proteins

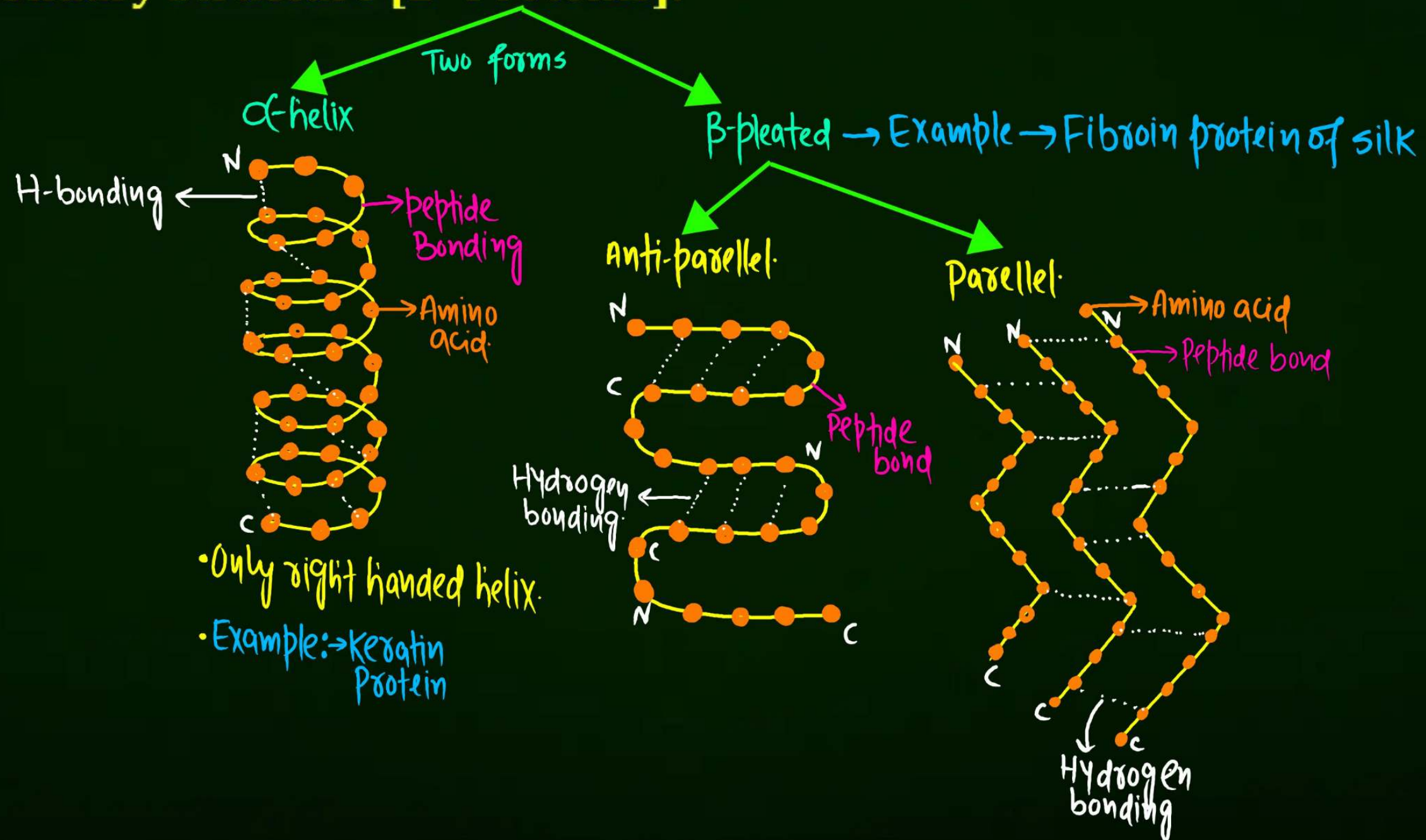
1. Primary structure [1° Proteins]:



- primary structure represents a straight chain of amino acids which forms a protein.
- Only peptide bond is present
- It gives positional information of amino acids.
- Example: → Insulin chain

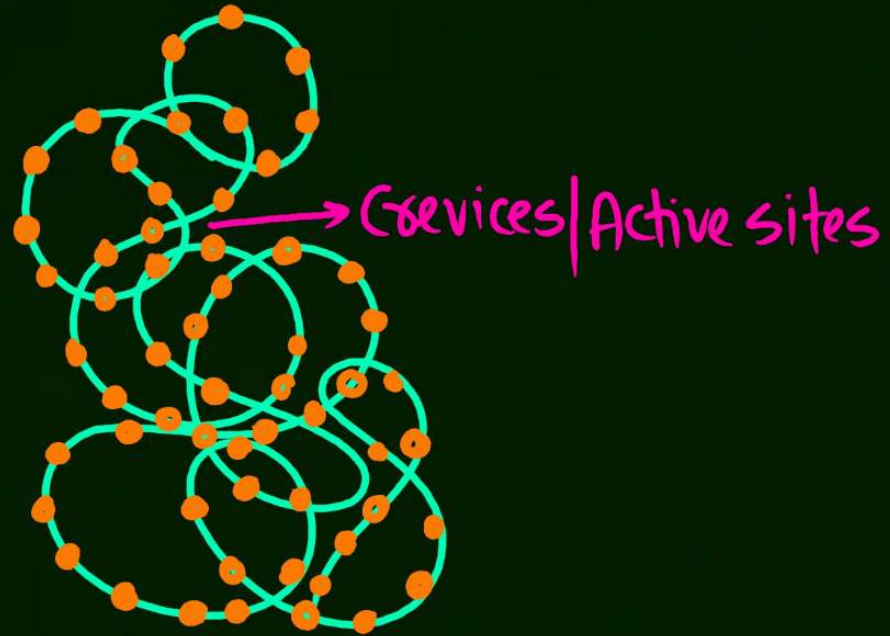
Structures of Proteins

2. Secondary Structure [2° Proteins]:



Structures of Proteins

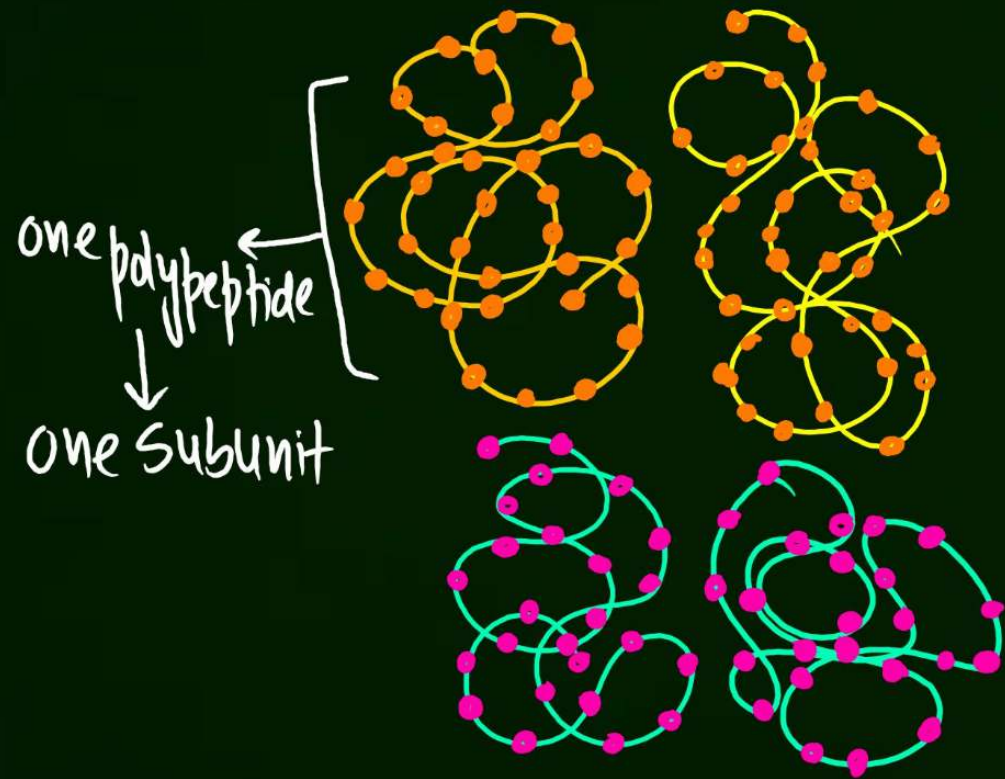
3. Tertiary (3°) Structure:



- A polypeptide chain is folded upon just like a wollen ball.
- 3°-protein form enzymes.
- When a polypeptide folded, crevices/grooves are formed called active sites.
- H-bonds, Disulphide bonds, Ionic bonds, peptide bonds are present.
- Example : → Myoglobin.

Structures of Proteins

4. Quaternary Structure. (4° Proteins):-



- 4° is the characteristic of more than one polypeptide
- Polypeptides sphere upon each other and form cube like.
- Example :-> Haemoglobin $\begin{cases} 2-\alpha\text{-chains} \\ 2-\beta\text{-chains} \end{cases}$

Functions of Proteins

- Collagen-protein → Intercellular ground substance
 - Trypsin → Enzyme
 - Insulin → Hormone
 - Antibody → Fights against infection.
 - Receptor → Sensory reception.
 - GLUT-4 → Enables Glucose transport into cells.
- ↓ Glucose ↓ Transfer.

Question: JKBOSE-2026

Alpha-helix is found in which level of protein structure?

- A** Quaternary structure
- B** Tertiary structure
- C** Primary structure
- D** Secondary structure



Question [NEET-2026]

Kaun Banayga Nabkaar



Match List I with List II.

List-I

- A. Trypsin
- B. Morphine
- C. Concanavalin A
- D. Collagen

List II

- I. Intercellular ground substance
- II. Lectin
- III. Enzyme
- IV. Alkaloid

Choose the correct answer from the options given below.

1 A-III, B-IV, C-II, D-I

2 A-I, B-II, C-III, D-IV

3 A-III, B-II, C-IV, D-I

4 A-IV, B-III, C-II, D-I

Question: JKBOSE-2025

Which among the following protein has Quaternary Structure?

A Myoglobin 3° X

B Haemoglobin

C Keratin $\rightarrow 2^\circ$ X

D All of These X



Thank
You